



Summary / Abstract

The Campus Complaint Management System (CCMS) is a modern, full-stack web application designed to streamline the reporting, tracking, and resolution of institutional grievances. It replaces fragmented, manual systems with a centralized digital platform. The system leverages Artificial Intelligence (Google Gemini API) to automatically classify complaints, integrates strict Service Level Agreement (SLA) policies to ensure timely resolutions, and provides specialized, role-based dashboards for students, staff, and administrators. This implementation results in a highly transparent, accountable, and efficient workflow for managing campus infrastructure and services.

Objective / Research Question

To develop a robust, secure, and user-friendly web application that automates the entire lifecycle of campus complaints. The core objectives are to reduce complaint resolution time, ensure staff accountability through automated SLA tracking and escalations, provide real-time status updates to submitters, and utilize AI to automatically parse and route issues to appropriate institutional departments without manual triage.

Method / Approach

The project was developed using a modern, scalable technology stack:

Frontend: Built with React 18 (Vite) and TypeScript. It utilizes Zustand for state management, React Query for data fetching, and Recharts for analytics. The UI employs a custom, premium “Editorial Glassmorphism” aesthetic with seamless light/dark mode support.

Backend: Powered by FastAPI (Python), serving a high-performance asynchronous REST API. It uses SQLAlchemy 2.0 (async) for ORM and Alembic for database migrations.

AI Integration: Integrated Google’s Gemini API via a dedicated `ai_service.py` to process complaint titles/descriptions and automatically return validated JSON categorization data.

Architecture & Logic: Implemented strict Role-Based Access Control (RBAC) separating Students, Staff, and Admins. Core business logic includes a dynamic SLA countdown system that monitors complaint deadlines based on severity (Low, Medium, High, Critical) and automatically flags overdue tasks.

Results / Findings

The implemented system successfully automates and modernizes institutional grievance redressal:

AI Categorization: The Gemini integration accurately extracts keywords and assigns relevant categories instantly upon submission, eliminating the need for manual dispatching.

SLA Enforcement: The SLA engine effectively calculates dynamic deadlines, visually alerting staff with color-coded badges and providing Admins with an “Escalated” dashboard for overdue items.

Analytics: The Admin Dashboard successfully processes live database metrics to generate visual reports on submission trends, category distributions, and staff resolution performance.

User Experience: The responsive glassmorphism UI provides high-contrast, visually engaging feedback, significantly reducing the friction traditionally associated with filing complaints.

DriveLink: [LINK](#) , [DIAGRAM](#)

Conclusion / Significance / Implications

CCMS modernizes campus administration by combining AI-driven complaint triage with strict SLA accountability to ensure issues are resolved promptly. This intelligent, transparent system proves that modern web technologies and AI can successfully automate and solve complex institutional workflows.